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Development and Validation of a Triplex RT-qPCR Assay for Rapid Clinical Diagnosis and Serotyping of Feline Infectious Peritonitis Virus

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Abstract

Feline infectious peritonitis (FIP) is a highly lethal disease caused by feline infectious peritonitis virus (FIPV), which poses significant diagnostic challenges in clinical practice. FIPV is divided into two serotypes (serotype I and serotype II) based on distinct serological responses driven by substantial sequence divergence in the spike (S) protein. Serotype I predominates in Europe and North America, whereas serotype II is more common in Asia. In this study, we developed a triplex reverse transcription quantitative PCR (RT-qPCR) assay for simultaneous detection and serotyping of FIPV. Primers and TaqMan probes were designed to target the conserved nucleocapsid (N) gene and serotype-specific regions within the S gene. After systematic optimization of reaction conditions, the final assay employed an annealing temperature of 64 °C and optimized primer–probe concentrations. The assay exhibited excellent linearity ($R^2 > 0.99$ for all targets), with amplification efficiencies ranging from 97.39% to 109.97%. No cross-reactivity was observed with other common feline pathogens, confirming high specificity. The limit of detection was as low as 0.5 copies/ μ L, and intra-assay repeatability showed coefficients of variation below 2.1%. Clinical validation using 63 feline samples revealed an overall FIPV positivity rate of 21.63%, with serotype II (17.46%) markedly more prevalent than serotype I (3.17%). Collectively, this triplex RT-qPCR assay demonstrates high sensitivity, exceptional specificity, and robust reproducibility, making it a valuable tool for rapid clinical diagnosis through the simultaneous detection of feline coronavirus (FCoV) and serotyping of FIPV.

Keywords: serotyping; feline infectious peritonitis virus; triplex RT-qPCR; identification



Academic Editor: Won Jong Rhee

Received: 22 January 2026

Revised: 22 February 2026

Accepted: 24 February 2026

Published: 26 February 2026

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1. Introduction

Feline infectious peritonitis (FIP) is a fatal systemic disease caused by feline infectious peritonitis virus (FIPV), a virulent biotype of feline coronavirus (FCoV). Characterized by extremely high mortality and complex immune-mediated pathogenesis, FIP poses significant diagnostic challenges in clinical practice [1]. FIPV exhibits strong tropism for monocytes and macrophages, leading to dysregulated cytokine responses and widespread vasculitis that manifest as effusive, non-effusive, or mixed clinical forms [2]. Current

diagnosis relies on a combination of clinical signs, ascitic fluid analysis, histopathology, and immunohistochemistry; however, these approaches are often invasive, time-consuming, and require specialized expertise, limiting their utility in early intervention [3].

FIPV is classified into two serotypes (serotype I and serotype II) based on substantial genetic divergence in the spike (S) glycoprotein gene, which results in distinct antigenic profiles and biological behaviors [4]. Serotype I is predominant globally, accounting for 80–95% of natural FIP cases, particularly in Europe and North America, whereas serotype II is more frequently reported in Asia and is thought to arise from recombination between FCoV and canine coronavirus [5,6]. The marked sequence variation in the S1 subunit of the S protein underlies serotype-specific neutralizing epitopes, rendering conventional serological assays prone to cross-reactivity and inaccurate serotyping [7].

The virus neutralization test (VNT) has historically served as the gold standard for FIPV detection but suffers from major drawbacks: it requires live-cell culture, takes 5–7 days to yield results, and cannot differentiate pathogenic FIPV from non-pathogenic feline enteric coronavirus (FECV), which share nearly identical genomes except for key mutations in the S and 3c genes [8]. While RT-PCR targeting the conserved 3' untranslated region (UTR) of FCoV enables sensitive viral screening, it lacks the resolution to distinguish FIPV from FECV or to assign serotype [9]. Recent efforts have focused on S gene-based molecular assays to improve specificity; however, most remain singleplex or lack robust validation for dual-serotype discrimination in clinical settings [10].

Notably, advances in multiplex real-time RT-qPCR have demonstrated great promise for the rapid and accurate detection of veterinary viral pathogens. For instance, triplex or quadruplex assays have been successfully developed for porcine reproductive and respiratory syndrome virus (PRRSV), avian influenza, and canine parvovirus, offering high throughput and cost-effective surveillance [11]. In the context of feline coronaviruses, a recent study established a duplex RT-qPCR targeting the M and N genes for FCoV load quantification, highlighting the feasibility of molecular tools in FIP diagnostics [12]. Another study emphasized the critical role of S protein variability in FIPV pathogenesis and proposed S gene sequencing as a complementary diagnostic strategy [13]. Furthermore, a recent review underscored the urgent need for serotype-discriminatory assays to support vaccine development and the epidemiological tracking of emerging FCoV variants [14]. While distinguishing pathogenic FIPV from non-pathogenic FECV remains a fundamental challenge in FCoV diagnostics, current molecular approaches primarily focus on viral detection and serotype discrimination due to the extensive genomic homology between the two biotypes [8,14]. The nucleocapsid (N) gene, being highly conserved across FCoV biotypes, serves as an ideal target for pan-FCoV detection but cannot differentiate between FIPV and FECV. Therefore, the primary objective of this study is to develop a triplex RT-qPCR assay capable of simultaneous FCoV detection and serotype discrimination, addressing the critical need for rapid serotyping in clinical and epidemiological settings.

To address this unmet need, a triplex real-time RT-qPCR assay capable of the simultaneous detection and serotyping of FIPV was developed in this study. By designing TaqMan probes targeting the conserved nucleocapsid (N) gene for pan-FIPV detection and serotype-specific regions within the S gene for differentiation of serotypes I and II, this method achieved high sensitivity (as low as 0.5 copies/ μ L), excellent specificity (no cross-reactivity with common feline pathogens), and outstanding reproducibility (intra-assay coefficient of variation < 2.10%). Clinical validation using 63 feline samples revealed an overall FIPV positivity rate of 21.63%, with serotype II (17.46%) significantly more prevalent than serotype I (3.17%) in the tested cohort. This assay provides a rapid, reliable, and scalable molecular diagnostic platform that enhances early FIP detection, supports precise serotype surveillance, and facilitates evidence-based control strategies in feline populations.

2. Results

2.1. Development and Optimization of a Triplex RT-qPCR Detection Platform

2.1.1. Systematic Validation of Individual Primer–Probe Combinatorial Specificity

To evaluate the functional performance of each primer–probe set within the triplex detection architecture, individual monoplex qPCR reactions were conducted using three distinct standard plasmids, each containing the target sequence corresponding to FIPV-N, FIPV-I-S, or FIPV-II-S. Notably, each reaction incorporated the full triplex primer–probe pool to rigorously assess potential cross-reactivity while verifying target-specific amplification.

The results demonstrated that clear, characteristic amplification curves, with distinct threshold cycle (CT) values, were generated only when the cognate plasmid (i.e., the one harboring the matching target sequence) was used as the template (Table 1). Critically, no detectable amplification signals were observed with non-cognate plasmids, confirming the high specificity of each primer–probe combination and the absence of cross-talk or competitive interference within the multiplex reaction matrix.

Table 1. Experimental validation data for three target-specific primer–probe configurations.

Templates	Primer–Probe Sets		
	FIPV-N-F/R/P ¹	FIPV-I-S-F/R/P	FIPV-II-S-F/R/P
FIPV-N plasmid	26.56 ²	—	—
FIPV-I-S plasmid	—	25.17	—
FIPV-II-S plasmid	—	—	28.34
Mixed Plasmids ³	26.05	22.15	27.17

¹ F/R/P denotes the combinatorial configuration of forward primer, reverse primer, and hydrolysis probe.

² Numerical values represent threshold cycle (Ct) values derived from qPCR amplification kinetics; the presence of a minus sign (—) indicates absence of detectable amplification within the reaction matrix. ³ The composite plasmid mixture was prepared through equimolar combination of three individual plasmids harboring target sequences corresponding to FIPV-N, FIPV-I-S, and FIPV-II-S genomic regions.

These findings validate that the engineered primer–probe sets retain both analytical sensitivity and precise target discrimination even when co-present in a consolidated reaction system, thereby establishing a robust technical foundation for the subsequent optimization and implementation of the triplex qPCR platform.

2.1.2. Optimization of Primer–Probe Concentration Parameters

To determine the optimal working concentrations for each primer–probe pair, a comprehensive titration strategy was implemented with systematic concentration gradients. Forward and reverse primers designed against conserved genomic regions of the target pathogen were evaluated across a spectrum of final concentrations (100, 150, 200, 250, and 300 nM), while corresponding hydrolysis probes were assessed at 50, 100, 150, 200, and 250 nM. Throughout this optimization phase, all other reaction components maintained constant concentrations to eliminate confounding variables and ensure methodological rigor. Amplification experiments utilized plasmid constructs containing target sequences as template substrates under strictly controlled conditions. Performance evaluation integrated dual analytical parameters: mean threshold cycle (Ct) values and absolute fluorescence intensity measurements (Rn). Figure 1a,c presents the mean Ct values for three primer pairs and three probes, respectively, across the concentration spectrum, while corresponding Rn intensity profiles are illustrated in Figure 1b,d. Optimal concentration selection prioritized parameter combinations yielding minimal Ct values, indicative of superior amplification kinetics. In scenarios where multiple concentrations produced comparable Ct values, the selection criterion shifted to maximize Rn signal intensity, reflecting enhanced detection sensitivity and assay robustness. This systematic analytical framework enabled

precise determination of optimal working concentrations for all primer–probe combinations, which were subsequently implemented in downstream validation studies and clinical diagnostic applications.

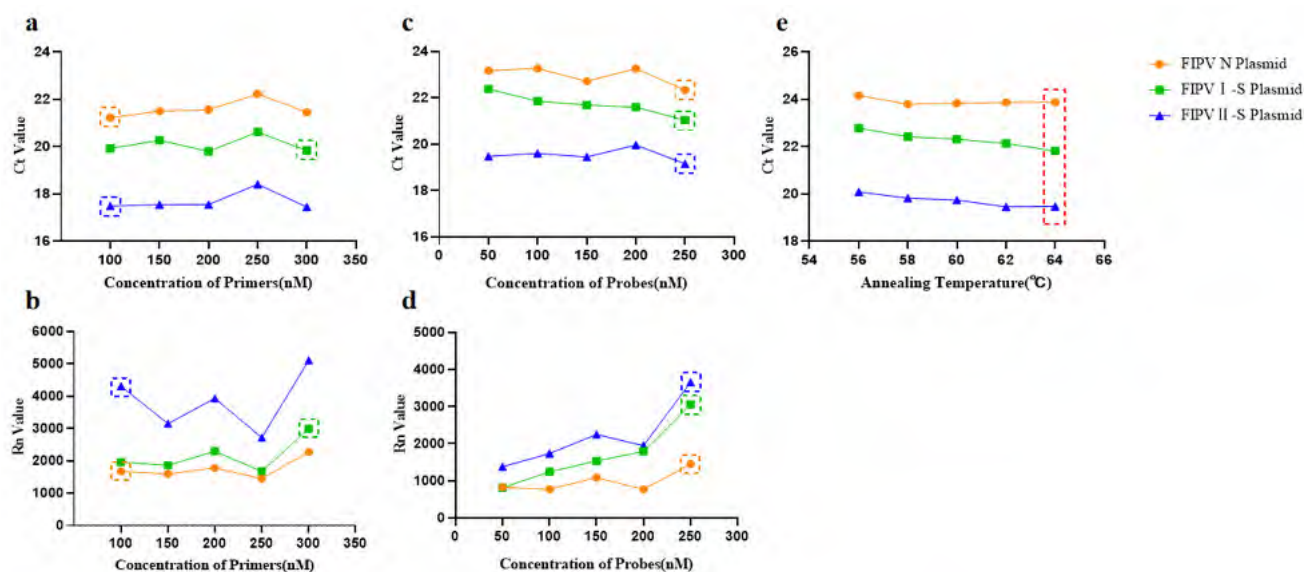


Figure 1. Systematic optimization of the triplex qPCR detection platform. (a) Mean threshold cycle (Ct) values for three primer pair configurations across varying forward and reverse primer concentrations. (b) Corresponding normalized reporter fluorescence (Rn) intensity profiles at each primer concentration gradient. (c) Mean Ct values for three hydrolysis probe concentrations evaluated against all target sequences. (d) Associated Rn signal intensities at each probe concentration parameter. (e) Comparative Ct value profiles across a thermal gradient of annealing temperatures (56–64 °C). Optimal concentration parameters and thermal conditions are highlighted within dashed rectangular boundaries, indicating maximal amplification efficiency and fluorescence signal generation.

2.1.3. Optimal Annealing Temperature Determination for Triplex Amplification

Following comprehensive optimization of primer and probe concentrations, thermal parameter refinement was systematically performed across a gradient of annealing temperatures (56 °C, 58 °C, 60 °C, 62 °C, and 64 °C). Standard plasmid constructs containing the target genomic sequences were precisely diluted to 1×10^5 copies/ μ L and used as quantification templates under strictly controlled experimental conditions. The optimal annealing temperature was empirically identified based on the lowest threshold cycle (CT) values, which directly reflect maximal amplification kinetics and reaction efficiency. Critical evaluation revealed that 64 °C yielded the best performance: it produced significantly lower CT values across all three targets and exhibited exceptional uniformity in amplification curve morphology and fluorescence kinetics (Figure 1e). This temperature strikes an optimal balance between primer annealing specificity and polymerase processivity, thereby maximizing multiplex detection sensitivity while preserving stringent target discrimination, which is essential for accurate serotype differentiation in complex clinical samples.

2.1.4. Quantitative Calibration Curve Development and Amplification Efficiency Assessment

To rigorously assess the amplification kinetics and linear dynamic range of the developed triplex qPCR assay, quantitative calibration curves were generated under the optimized reaction conditions. A 10-fold serial dilution series of recombinant plasmid DNA containing the target genomic sequences was prepared, spanning a concentration range from 1×10^8 to 1×10^3 copies/ μ L. Standard curves were constructed by plotting threshold cycle (CT) values against the log₁₀-transformed template concentrations, as

shown in Figure 2a–c. The resulting calibration curves exhibited excellent linearity, with slopes of -3.104 (corresponding to 109.97% amplification efficiency), -3.386 (97.39%), and -3.284 (101.61%) for the FIPV-N, FIPV-I-S, and FIPV-II-S targets, respectively. All correlation coefficients (R^2) exceeded 0.99, confirming robust linear relationships and consistent amplification kinetics across the entire dynamic range. The corresponding amplification plots for each target are displayed in Figure 2d–f. Collectively, these quantitative validation data demonstrate that the triplex platform delivers high-performance amplification for all three genomic targets, combining exceptional analytical sensitivity with reliable quantification accuracy over a broad concentration span. These characteristics underscore its suitability for precise viral load measurement in complex clinical specimens.

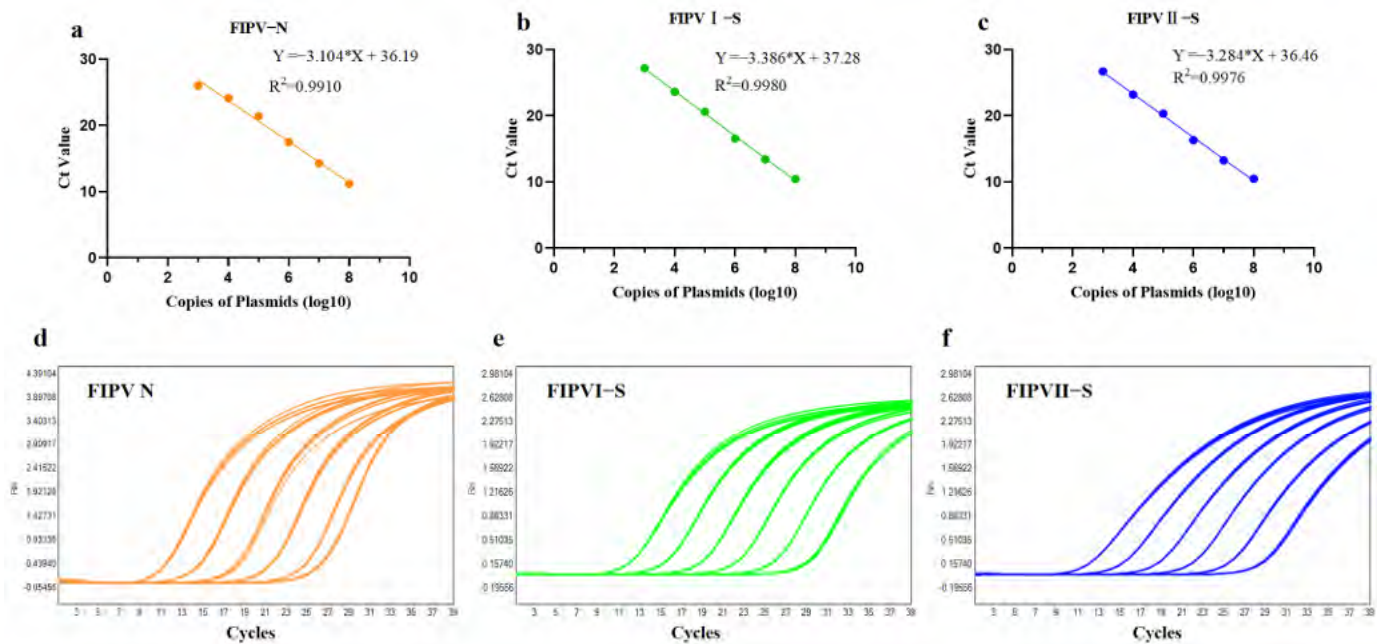


Figure 2. Quantitative calibration profiling for three target-specific plasmid constructs. (a–c) Standard calibration curves generated through systematic serial dilution of recombinant plasmid DNA across a dynamic concentration range. (d–f) Corresponding fluorescence amplification kinetics profiles for FIPV-N, FIPV-I-S, and FIPV-II-S genomic targets, respectively. All experimental measurements were performed in biological triplicate to ensure statistical robustness. The asterisk symbol (*) denotes the multiplication operator within the derived linear regression equations for quantitative analysis.

2.2. Analytical Specificity Profiling and Cross-Reactivity Assessment

To rigorously evaluate the target specificity of the developed triplex qPCR assay, recombinant plasmid standards containing the target gene sequences were precisely diluted to 1×10^5 copies/ μ L and used as primary templates. In addition, comprehensive cross-reactivity testing was performed against genomic nucleic acids from two FIPV serotype II reference strains (FIPV DF2 and FIPV 79-1146) and six other feline infectious pathogens archived in our institutional biorepository: feline parvovirus (FPV), feline calicivirus (FCV), feline herpesvirus (FHV), feline leukemia virus (FeLV), feline immunodeficiency virus (FIV), and feline rotavirus (FRV).

Critical analysis of the amplification profiles revealed exceptional target discrimination: robust and specific signals were detected only for the intended cognate targets. No amplification was observed with any non-target pathogens or with host genomic DNA extracted from feline blood samples, confirming the high analytical specificity of the assay (Table 2). This stringent specificity profile underscores the reliability of the platform for

accurate differential diagnosis in complex clinical specimens that may harbor multiple co-circulating feline pathogens.

Table 2. Analytical specificity profile of the triplex qPCR detection platform.

DNA/RNA Templates	Targets		
	FIPV-N	FIPV-I-S	FIPV-II-S
FIPV-N plasmid	+ ¹	—	—
FIPV-I-S plasmid	—	+	—
FIPV-II-S plasmid	—	—	+
FIPV DF2	+	—	+
FIPV 79-1146	+	—	+
FPV	—	—	—
FCV	—	—	—
FHV	—	—	—
FeLV	—	—	—
FIV	—	—	—
FRV	—	—	—

¹ “+” indicates positive amplification with definitive fluorescence threshold crossing; “—” denotes the absence of detectable amplification above the established background baseline throughout the complete thermal cycling regimen.

2.3. Analytical Sensitivity Assessment and Detection Limit Determination

To rigorously assess the analytical sensitivity of the triplex qPCR assay, recombinant standard plasmids were serially diluted to precisely defined concentrations of 10, 5, 1, and 0.5 copies/μL. An equimolar mixture of all three plasmid constructs was then prepared to serve as a comprehensive template matrix mimicking a multiplex clinical scenario. Experimental validation demonstrated exceptional detection performance: at the 5 copies/μL concentration, 100% of replicates produced clear and reproducible amplification signals in FIPV-I-S; at the 0.5 copies/μL concentration, 100% of replicates produced clear and reproducible amplification signals in FIPV-N and FIPV-II-S (Table 3). Based on this reproducibility threshold, as low as 0.5 copies/μL was established as the validated limit of detection (LOD) for the simultaneous identification of all three genomic targets, confirming the assay’s high analytical sensitivity. Representative amplification curves from the sensitivity assessment are shown in Figure 3. Collectively, these validation data underscore the clinical utility of this platform for reliably detecting the low-abundance viral targets commonly encountered in real-world diagnostic specimens, thereby supporting its application in the early and accurate diagnosis of feline infectious diseases.

Table 3. Analytical sensitivity profiling of the triplex qPCR detection platform.

Templates	Positive Rates (%)			
	10 Copies/μL	5 Copies/μL	1 Copies/μL	0.5 Copies/μL
FIPV-N	100 ¹	100	100	100
FIPV-I-S	100	100	33	25
FIPV-II-S	100	100	100	100

¹ Each experimental condition was evaluated through 12 independent technical replicates to ensure statistical robustness and analytical reproducibility.

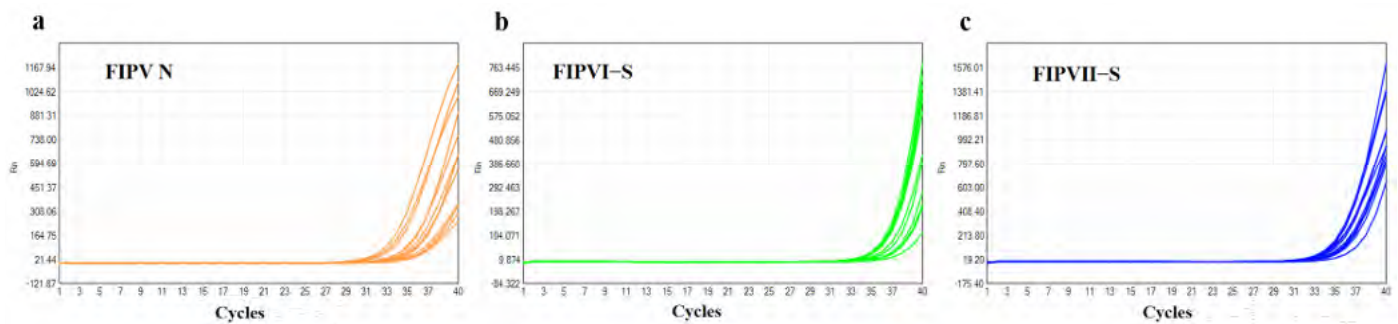


Figure 3. Representative amplification profiles generated during analytical sensitivity assessment at 5 copies/ μL threshold concentration. The figure displays detection outcomes for (a) FIPV-N, (b) FIPV-I-S, and (c) FIPV-II-S. Each analytical condition was subjected to 12 independent technical replicates to confirm detection reliability and assay robustness under limiting template conditions.

2.4. Analytical Repeatability Assessment and Inter-Assay Precision Validation

To comprehensively evaluate the analytical precision and reproducibility of the developed triplex qPCR platform, coefficient of variation (CV) analyses were performed on threshold cycle (CT) values to systematically assess both intra-assay and inter-assay variability. Standard plasmid constructs at precisely calibrated concentrations of 1×10^5 and 1×10^2 copies/ μL were used as quantitative templates throughout the validation. For intra-assay precision, each concentration was tested in 20 technical replicates within a single run; for inter-assay reproducibility, the same measurements were repeated across three independent experimental runs conducted on separate days. As shown in Table 4, the mean CT values exhibited remarkably low dispersion across all replicates. Standard deviations remained consistently minimal, ranging from 0.33 to 0.62, reflecting exceptional assay stability and detection reliability. Notably, both the intra- and inter-assay CVs were consistently below 2.1% for all target sequences and concentration levels, demonstrating outstanding methodological robustness under varying experimental conditions. Representative amplification profiles illustrating this high level of precision are presented in Figure 4. Collectively, these rigorous validation data confirm that the triplex qPCR platform delivers highly consistent, reproducible, and precise quantitative results, making it exceptionally well-suited for routine diagnostic use and reliable viral load monitoring in clinical management protocols for feline infectious peritonitis.

Table 4. Reproducibility performance metrics of the triplex qPCR detection platform.

Templates	Copies of Plasmids (Copies/ μL)	Ct Value ¹	CI ²	CVs (%)
FIPV-N	10^5	21.09 ± 0.33	[20.93, 21.25]	1.55%
FIPV-I-S	10^5	20.28 ± 0.35	[20.11, 20.45]	1.74%
FIPV-II-S	10^5	20.00 ± 0.41	[19.80, 20.20]	2.07%
FIPV-N	10^2	31.27 ± 0.54	[31.00, 31.54]	1.73%
FIPV-I-S	10^2	30.60 ± 0.52	[30.34, 30.86]	1.71%
FIPV-II-S	10^2	30.02 ± 0.62	[29.71, 30.33]	2.06%

¹ Mean threshold cycle (Ct) values and associated standard deviations (SD) were systematically calculated and integrated into the tabular dataset presentation. ² The 95% confidence intervals (CI) were statistically determined, with numerical values positioned before parentheses denoting the lower confidence boundary and values following parentheses representing the upper confidence boundary.

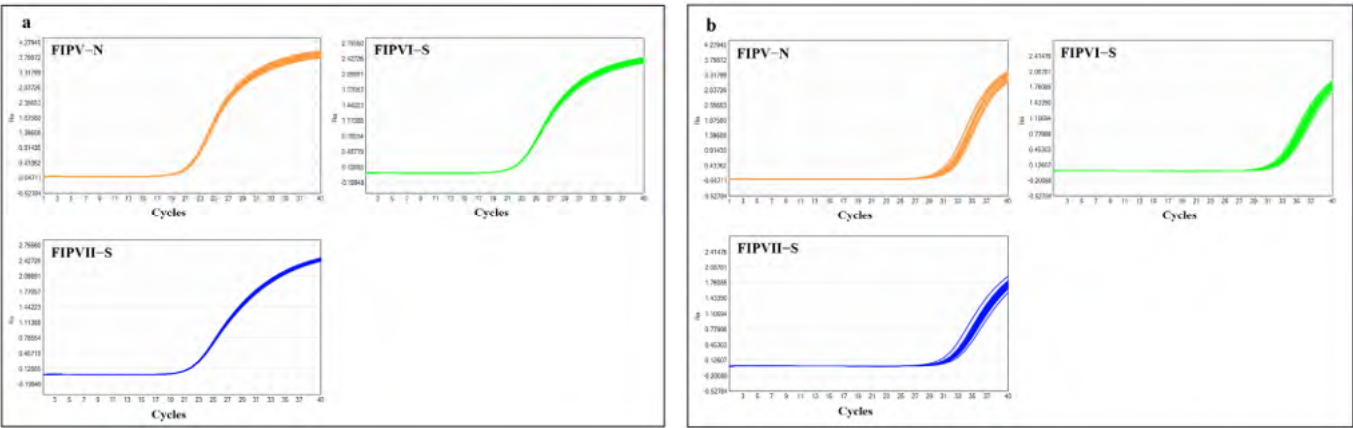


Figure 4. Amplification profiles derived from analytical repeatability validation experiments. The panel illustrates detection performance for (a) high-concentration plasmid template (1 × 10⁵ copies/μL) and (b) low-concentration plasmid template (1 × 10² copies/μL). Each template concentration underwent 20 independent technical replicates within a single assay run to rigorously evaluate intra-assay precision and detection consistency under standardized thermal cycling conditions.

2.5. Clinical Diagnostic Validation and Field Specimen Assessment

To critically assess the clinical utility and diagnostic performance of the established triplex qPCR platform, 63 clinical specimens were collected from companion animal hospitals across Beijing and Jiangsu Province. The sample cohort encompassed a diverse array of specimen types, including oropharyngeal swabs, nasal secretions, rectal swabs, and ascitic fluid. Application of the validated assay identified 13 FIPV-positive cases, yielding a clinical detection rate of 21.63%, with threshold cycle (CT) values ranging from 21 to 34. Serotype discrimination revealed that two of these positive cases (15.38% of positives) belonged to FIPV serotype I, while the remaining 11 (84.62%) were classified as serotype II (Table 5). This serotype distribution closely mirrors regional epidemiological patterns previously reported in Asian feline populations, reinforcing the necessity of serotype-specific molecular diagnostics for accurate disease surveillance, informed clinical decision-making, and targeted therapeutic strategies in veterinary practice. To validate the diagnostic accuracy of the developed assay, parallel testing was performed using a previously published TaqMan qPCR method targeting the FIPV-N gene as the reference standard [15]. All clinical specimens were re-evaluated with this reference method. The results demonstrated that the established triplex RT-qPCR assay achieved a sensitivity of 92.31%, a specificity of 98.00%, and an overall agreement of 96.83% (Table 6).

Table 5. Clinical diagnostic performance metrics of the triplex qPCR detection platform.

Target Detection (Fluorescence Signal)	Number of Positive Samples/Total Samples (Positive Rates%)
FIPV-N (ROX)	13/63 (21.63%)
FIPV-I-S (VIC)	2/63 (3.17%)
FIPV-II-S (FAM)	11/63 (17.46%)

Table 6. Comparison of the developed triplex qPCR assay and reference methods.

Reference Method	Kappa Test	Developed Method		Total	Sensitivity (%)	Specificity (%)	Agreement (%)
		+	−				
[15]	+	12	1	13	92.31	98.00	96.83
	−	1	49	50			
	Total	13	50	63			

3. Discussion

Feline infectious peritonitis virus (FIPV), a highly pathogenic mutant derived from feline coronavirus (FCoV), has emerged as a major infectious threat to domestic cat populations in China over the past decade [1]. Characterized by high mortality, particularly among kittens and intact cats, and substantial economic burden due to prolonged therapeutic regimens, FIP poses significant challenges to both pet owners and commercial catteries [7]. Current control strategies in China remain hampered by multiple constraints: reliance on insensitive or non-specific diagnostic tools, lack of nationally approved antivirals or vaccines, and insufficient surveillance infrastructure coupled with a dearth of regionally representative viral strain repositories [16].

Although recent advances in antiviral therapeutics (e.g., GS-441524) have shown clinical promise [17], their unregulated use underscores the urgent need for accurate, early diagnosis to guide rational intervention. Moreover, while next-generation vaccine platforms, including mRNA-based constructs targeting the receptor-binding domain (RBD) of the FIPV spike (S) protein, offer theoretical advantages in safety and immunogenicity, they remain largely experimental, with unresolved issues regarding delivery efficiency, cold-chain logistics, and cost-effectiveness in veterinary settings [18,19]. Critically, the existence of two genetically and antigenically distinct FIPV serotypes, serotype I (predominant in Asia) and serotype II (arising from recombination between FCoV and canine coronavirus S genes), further complicates vaccine design and deployment [20,21]. These serotypes exhibit divergent epidemiological distributions and potentially differential pathogenic mechanisms, rendering universal vaccine approaches ineffective without precise serotype identification [22].

In this context, molecular diagnostics capable of simultaneous detection and serotyping are essential for effective disease surveillance, outbreak response, and tailored clinical management. Our assay employs a dual-targeting strategy: the highly conserved nucleocapsid (N) gene enables pan-FIPV detection across both serotypes, while hypervariable yet intra-serotype-conserved regions within the S gene allow unambiguous differentiation of FIPV-I and FIPV-II. This design deliberately avoids mutation-prone domains, thereby enhancing long-term assay robustness against viral evolution, a critical consideration given FCoV's high recombination and mutation rates [23]. Through systematic optimization of primer–probe concentrations and annealing temperature (64 °C), we effectively minimized competitive amplification and cross-talk, common pitfalls in multiplex systems [24].

The resulting platform demonstrates exceptional analytical performance: amplification efficiencies of 97.39–109.97%, linearity ($R^2 > 0.99$) across a six-log dynamic range (10^8 to 10^3 copies/ μ L), and a limit of detection (LOD) as low as 0.5 copies/ μ L, representing a ~20-fold improvement over conventional RT-PCR (~100 copies/ μ L) [25]. Notably, intra- and inter-assay coefficients of variation remained below 2.1%, confirming high reproducibility suitable for routine diagnostics. When applied to 63 prospectively collected clinical specimens from Beijing and Jiangsu, the assay identified FIPV in 21.6% of samples, with serotype II accounting for 84.6% of positives, consistent with regional epidemiological reports from East Asia [21,26]. The ability to rapidly distinguish between serotypes not only informs

prognosis (as some evidence suggests differential virulence [27]) but also supports future vaccine efficacy trials that must account for serotype-specific immune responses.

Current diagnostic approaches for feline infectious peritonitis virus (FIPV) primarily rely on serological assays, virus neutralization testing (VNT), and conventional RT-PCR. While enzyme-linked immunosorbent assays (ELISAs) or indirect immunofluorescence offer operational simplicity, they suffer from a fundamental inability to distinguish pathogenic FIPV from its benign counterpart, feline enteric coronavirus (FECV). Both viruses share extensive antigenic homology, resulting in substantial cross-reactivity, particularly between FIPV serotypes I and II, with reported false-positive rates ranging from 40% to 60% in endemic regions [28,29]. Although VNT is widely regarded as the serological gold standard due to its ability to differentiate neutralizing antibody responses, its requirement for live-cell culture and prolonged incubation (5–7 days) renders it unsuitable for timely clinical decision-making [30].

In contrast, the triplex RT-qPCR platform developed in this study overcomes these critical limitations through a molecularly precise, rapid, and integrated diagnostic strategy. Our assay demonstrates absolute analytical specificity: no cross-reactivity was observed against a comprehensive panel of common feline pathogens, including feline parvovirus (FPV), feline calicivirus (FCV), feline herpesvirus (FHV), feline leukemia virus (FeLV), feline immunodeficiency virus (FIV), and feline rotavirus (FRV), nor against host genomic DNA from healthy cats (Table 4). This level of discrimination is essential in multi-pathogen clinical settings where co-infections are frequent.

Moreover, the assay achieves remarkable time efficiency, delivering complete results—including detection and serotype identification—in under two hours. This represents a >95% reduction in turnaround time compared to VNT protocols, enabling same-day diagnosis and facilitating early therapeutic intervention, a key determinant of survival in effusive FIP cases [17]. When applied to 63 prospectively collected clinical specimens, the platform identified FIPV in 21.6% of samples, with serotype II accounting for 84.6% of positives (17.46% of the total cohort) versus only 3.17% for serotype I. This distribution strongly aligns with epidemiological data from East Asia, where FIPV-I is genetically dominant but FIPV-II remains clinically relevant in specific contexts such as cattery outbreaks or interspecies transmission events [21,26].

While emerging technologies such as digital PCR (dPCR) have demonstrated exceptional sensitivity (down to 1 copy/ μ L) for FIPV detection [31], their widespread adoption in veterinary diagnostics is hindered by high instrumentation costs, limited throughput, and complex workflow requirements. In contrast, our triplex RT-qPCR system strikes an optimal balance between high analytical performance, exhibiting a limit of detection as low as 0.5 copies/ μ L, and practical feasibility. It operates on widely available real-time PCR platforms (e.g., SLAN-96S), uses cost-effective reagents, and requires minimal technical expertise, making it highly suitable for routine implementation in diagnostic laboratories and companion animal hospitals.

Importantly, the dual-target design, leveraging conserved N gene regions for pan-FIPV detection and serotype-specific S gene segments for discrimination, ensures both robustness against viral genetic drift and precision in epidemiological tracking. This is particularly valuable given the high recombination rate of alphacoronaviruses such as FCoV, which can rapidly generate novel variants that evade single-epitope-based assays [23,32].

Collectively, our platform addresses the core gaps in current FIPV diagnostics: lack of specificity, slow turnaround, absence of serotyping capability, and poor scalability. By providing a rapid, accurate, and economically viable solution, it lays the groundwork for improved disease surveillance, informed vaccine deployment (as serotype-mismatched

vaccines may offer limited cross-protection [22]), and better clinical outcomes in affected feline populations.

The triplex RT-qPCR platform developed in this study was prospectively applied to 63 clinical specimens collected from companion animal hospitals across Beijing and Jiangsu Province. Among the 13 FIPV-positive cases identified, serotype II accounted for 84.6% (11/13), while serotype I represented only 15.4% (2/13). This striking predominance of serotype II aligns with, but also refines, existing epidemiological models in Asia, where serotype I is generally considered dominant in natural infections [21,26]. The higher-than-expected detection of serotype II in northern and eastern China may reflect localized viral ecology, intensive multi-cat housing practices, or interspecies transmission dynamics involving canine coronavirus (CCoV), given that FIPV serotype II arose through recombination between FCoV/FIPV-I and CCoV S gene sequences [33,34].

This serotype-specific distribution carries profound implications across three critical domains of feline health management:

First, vaccine selection and efficacy. The only commercially available modified-live FIP vaccine (Primucell® FIP, Zoetis, Parsippany, NJ, USA) is derived from an FIPV-II strain (79-1146) but demonstrates limited cross-protection against field-isolated FIPV-I due to substantial antigenic divergence in the spike (S) protein receptor-binding domain (RBD) [27]. Conversely, emerging subunit or mRNA vaccines based on FIPV-I RBD may offer suboptimal protection against serotype II [18,19]. Thus, precise serotyping—enabled by our assay—is essential for rational vaccine deployment and interpreting vaccine trial outcomes in region-specific contexts.

Second, antiviral therapy optimization. Recent pharmacological studies have revealed serotype-dependent differences in drug susceptibility. Pedersen et al. (2022) demonstrated that FIPV-II exhibits approximately 1.8-fold greater sensitivity to the nucleoside analog GS-441524 ($EC_{50} \approx 0.38 \mu\text{M}$) compared to FIPV-I ($EC_{50} \approx 0.69 \mu\text{M}$), likely due to structural variations in the viral RNA-dependent RNA polymerase (RdRp) [8]. Such findings underscore the potential for serotype-informed dosing regimens to enhance therapeutic efficacy while minimizing drug resistance—a paradigm shift toward precision veterinary medicine.

Collectively, our findings demonstrate that serotype-resolved molecular diagnostics are not merely an academic exercise but a clinical and public health imperative. By delivering rapid, accurate, and cost-effective serotyping within a single reaction, the triplex RT-qPCR platform described here provides a scalable solution for real-world implementation in veterinary practice, research, and surveillance systems alike.

4. Materials and Methods

4.1. Viral Isolates, Nucleic Acids, and Clinical Specimens

FIPV serotype II reference strains FIPV 79-1146 (GenBank accession DQ010921.1) and FIPV DF2 (GenBank accession DQ286389.1) were maintained in our laboratory. Nucleic acid extracts from feline pathogens, including feline parvovirus (FPV), feline calicivirus (FCV), feline herpesvirus (FHV), feline leukemia virus (FeLV), feline immunodeficiency virus (FIV), and feline rotavirus (FRV), were previously isolated and maintained in our laboratory. A total of 63 clinical specimens comprising ascitic fluid collections and rectal swab samples were prospectively obtained from collaborating veterinary practices in Beijing and Jiangsu Province. All biological materials were systematically cataloged and stored under cryopreservation conditions at -80°C in our dedicated sample repository facility.

4.2. Design and Synthesis of Serotype-Discriminating Primers and Hydrolysis Probes

Genomic sequences of feline infectious peritonitis virus (FIPV) isolated within China during the past quinquennium ($n = 69$) were retrieved from the NCBI GenBank public

repository. The nucleocapsid (N) and spike (S) protein-coding regions were systematically extracted from these genomic datasets. Comprehensive multiple sequence alignments were performed independently for each gene using the MUSCLE algorithm implemented within the MEGA11 bioinformatics platform. For universal FIPV detection, primer–probe sets (designated FIPV-N-F/R/P) were strategically designed targeting evolutionarily conserved genomic regions within the N gene that demonstrated complete sequence homology across both serotypes. Conversely, for serotype-specific discrimination, distinct primer–probe combinations (FIPV-I-S-F/R/P and FIPV-II-S-F/R/P) were engineered to target hypervariable regions within the S gene that exhibited intra-serotype conservation while maintaining inter-serotype divergence (Supplementary Figure S1). The complete oligonucleotide sequences and structural specifications are detailed in Table 7. All molecular reagents were custom-synthesized and quality-controlled by Shanghai GeneRay Biotechnology Co., Ltd. (Shanghai, China) according to established oligonucleotide manufacturing standards.

Table 7. Serotype-specific oligonucleotide sets for triplex RT-qPCR detection of FIPV.

Oligonucleotide Sets	Target Gene	Sequences (5′-3′)	Genomic Position ²	Reference Viral Strain
FIPV-N-F ¹	FIPV	AACACACCTGGAAGAAAACCTGC	27,470–27,491	DQ010921.1
FIPV-N-R		CCATTGGCAACGAGATCACTAT	27,551–27,572	
FIPV-N-P		ROX-TTGTCACATCTCCCTT-MGB	27,496–27,511	
FIPV-I-S-F	FIPV-I	TGTTGCAGTACAAGCCGAATAC	22,260–22,281	MT444152.1
FIPV-I-S-R		TGCCATTGCAAACATACTTAGC	22,318–22,339	
FIPV-I-S-P		VIC-CAGATTCAAAGTAAACCTGT-MGB	22,285–22,304	
FIPV-II-S-F	FIPV-II	TAATTGCTTGTGGCCAGTGC	21,350–21,369	OQ311323.1
FIPV-II-S-R		AAGACACACCATTACATTGGCT	21,420–21,441	
FIPV-II-S-P		FAM-AAACTGTGCACCTTCAA-MGB	21,403–21,419	

¹ F denotes forward primer, R denotes reverse primer, and P denotes hydrolysis probe. ² Genomic coordinates correspond to nucleotide positions within the reference viral genome sequence (GenBank accession numbers provided).

4.3. Viral Nucleic Acid Purification and Reference Plasmid Engineering

Viral RNA was extracted from cultured viral isolates and clinical specimens, including ascitic fluid and rectal swabs, using the UE Body Fluid Viral DNA/RNA Mini-Prep Kit (Uelandy Inc., Cat. No. UE-MN-BF-VNA-250, Suzhou, China) in accordance with the manufacturer's instructions. The purified RNA was eluted in 50 µL of RNase-free deionized water and stored at −80 °C for subsequent RT-qPCR analysis.

Three standard plasmids were constructed by synthesizing target gene fragments corresponding to the three primer–probe sets and cloning them into the pUC57 vector backbone via restriction enzyme-mediated ligation. These recombinant plasmids, designated FIPV-N, FIPV-I-S, and FIPV-II-S, were custom-generated by Suzhou Gen Script Biotechnology Co., Ltd. (Suzhou, China). The purified plasmid constructs were resuspended in TE buffer (Sengen Biotech, Cat. No. B541019, Shanghai, China) to a final concentration of 4 µg/µL and stored at −20 °C for downstream applications.

4.4. Systematic Optimization of Triplex RT-qPCR Reaction Parameters

4.4.1. Validation of Individual Primer–Probe Performance and Triplex Pool Compatibility

Individual validation of each primer–probe set and assessment of their compatibility in a triplex format were carried out as follows. The three synthesized standard plasmids (FIPV-N, FIPV-I-S, and FIPV-II-S) were independently diluted to a concentration of 10⁴ copies/µL and subjected to monoplex RT-qPCR using the Hifair[®] III One Step RT-qPCR Probe Kit (Yeasten Biotechnology, Shanghai, China). Reactions were assembled strictly in accordance with the manufacturer's technical specifications, with 5 µL of plasmid template added to each 20 µL reaction mixture. Subsequently, all three primer–probe pairs were combined into a single multiplex primer pool, with final concentrations adjusted

to 200 nmol/L for each primer and 100 nmol/L for each probe. A 2 µL aliquot of this pooled primer–probe mix was then incorporated into the reaction system for simultaneous detection of the three standard plasmids. Amplification and fluorescence detection were performed on the Slan-96S real-time PCR platform (Shanghai Hongshi Medical Technology Co., Ltd., Shanghai, China). During thermal cycling, fluorescence signals from the ROX, VIC, and FAM channels were monitored concurrently. Following amplification, threshold cycle (CT) values and normalized reporter fluorescence (Rn) were carefully recorded for each channel to evaluate amplification efficiency, specificity, and signal intensity profiles.

4.4.2. Systematic Titration and Optimization of Primer–Probe Concentration Parameters in Triplex Reaction Pools

To determine the optimal concentration of the FIPV-N-F/R primer pair within the triplex reaction system, a systematic titration approach was employed. Primer concentrations were incrementally varied across a gradient of 100 to 300 nmol/L, while the concentration of the corresponding FIPV-N-P probe and all other primer–probe components in the pool were held constant. The complete reaction composition for each titration condition is provided in Table 8. Amplification was performed using the previously established RT-qPCR protocol, with threshold cycle (CT) values and normalized reporter fluorescence (Rn) carefully recorded for each detection channel. Following primer optimization, an analogous titration strategy was applied to the FIPV-N-P probe, evaluating its performance across a concentration range of 50 nmol/L to 250 nmol/L. This comprehensive optimization workflow was then systematically extended to all remaining primer–probe sets in the triplex panel. The goal was to establish balanced amplification efficiencies across all targets while minimizing competitive inhibition and signal interference within the multiplex reaction matrix.

Table 8. Triplex primer–probe concentration optimization reaction matrix.

Reagent	Volume per Reaction (µL)
2 × Hifair [®] III P buffer	10
Hifair [®] UH III Enzymes	1
FIPV-N-F/R	0.2/0.3/0.4/0.5/0.6 ²
FIPV-N-P	0.4
FIPV-I-S-F/R/P ¹	0.4
FIPV-II-S-F/R/P	0.4
Plasmid template	3
Nuclease-free water	Up to 20

¹ F/R/P denotes forward primer, reverse primer, and probe, respectively; all detection probes were incorporated into the reaction matrix at a standardized concentration of 10 µmol/L. ² Five distinct concentration gradients of FIPV-N-F/R primers were systematically evaluated.

4.4.3. Optimal Annealing Temperature Determination for Triplex Amplification

Following the determination of optimal primer and probe concentrations, the triplex primer–probe pool was reconstituted accordingly. The amplification system was then established using a composite standard plasmid mixture (containing equimolar amounts of FIPV-N, FIPV-I-S, and FIPV-II-S) at a concentration of 10⁵ copies/µL as the template. Each 20 µL reaction included 2 µL of the optimized primer–probe formulation. Real-time amplification was performed under a gradient of annealing temperatures (56 °C, 58 °C, 60 °C, 62 °C, and 64 °C) to evaluate thermal sensitivity across the multiplex system. Threshold cycle (CT) values were meticulously recorded for each fluorescence detection channel, enabling the identification of the optimal annealing temperature that maximized amplification efficiency and signal consistency across all three targets.

4.5. Quantitative Calibration Curve Development and Amplification Efficiency Assessment

Serial 10-fold dilutions of the three standard plasmids were prepared to generate a dynamic concentration range spanning from 10^8 to 10^2 copies/ μ L. Each dilution was analyzed in triplicate under the optimized reaction conditions, with threshold cycle (CT) values systematically recorded for each fluorescent detection channel.

Following amplification, amplification efficiencies for all three targets were calculated from the slopes of their respective standard curves, using the formula outlined in Molecular Cloning: A Laboratory Manual (4th Edition) [35]:

$$\text{Efficiency (\%)} = (10 - 1/\text{slope} - 1) \times 100.$$

This quantitative calibration strategy enabled a precise evaluation of key analytical performance parameters, including sensitivity, linearity, and efficiency, across the full dynamic range of detection.

4.6. Target Specificity, Inter-Assay Reproducibility, and Analytical Sensitivity Determination

Specificity was evaluated using not only plasmids containing the target genes but also genomic nucleic acids extracted from two FIPV serotype II reference strains, FIPV 79-1146 and FIPV DF2, as well as a panel of nucleic acids from common feline pathogens, including feline parvovirus (FPV), feline calicivirus (FCV), feline herpesvirus (FHV), feline leukemia virus (FeLV), feline immunodeficiency virus (FIV), and feline rotavirus (FRV). To assess reproducibility, plasmid standards at concentrations of 10^5 and 10^2 copies/ μ L were subjected to 20 independent RT-qPCR runs using the optimized triplex primer–probe pool formulation. The analytical sensitivity (limit of detection) for each primer–probe set was determined by testing serially diluted plasmid standards at concentrations of 10, 5, 1, and 0.5 copies/ μ L, with each dilution analyzed in 12 replicates under standardized reaction conditions to establish robust and statistically supported detection limits.

4.7. Diagnostic Performance Assessment and Validation Using Clinical Specimens

To validate the diagnostic performance of the assay for clinical use, a comprehensive evaluation was conducted using 63 feline clinical specimens, including oropharyngeal, nasal, and rectal swabs, as well as ascitic fluid, prospectively collected from companion animal hospitals in Beijing and Jiangsu Province. Each specimen was analyzed in parallel with laboratory-maintained FIPV 79-1146 nucleic acid, which served as the positive reference control. Viral RNA extraction was carried out using the standardized protocol described above, followed by real-time amplification and detection on the SLAN-96S molecular diagnostic platform (Shanghai Hongshi Medical Technology Co., Ltd., Shanghai, China). This enabled the simultaneous quantification and serotype discrimination of FIPV targets within a single reaction, demonstrating the assay's suitability for routine clinical diagnostics.

5. Conclusions

This study successfully established a triplex RT-qPCR platform capable of rapid, simultaneous FCoV detection and serotype discrimination. The assay exhibits exceptional analytical performance, characterized by absolute target specificity, high sensitivity (limit of detection: as low as 0.5 copies/ μ L), and outstanding reproducibility (intra- and inter-assay coefficients of variation < 2.1%). Clinical validation confirmed its utility for serotype-resolved molecular diagnostics in real-world settings. As a robust, cost-effective, and scalable molecular diagnostic tool, this platform represents a significant advance in precision veterinary medicine for FCoV management, enabling serotype-guided therapeutic decision-making and epidemiological surveillance.

6. Patents

A patent application has been submitted for this research, with the following numbers: application (CN202510920485.4), online public examination (CN120758674A), and authorization (CN120758674B).

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/ijms27052204/s1>.

Author Contributions: Conceptualization, R.X. and H.J.; methodology, R.X., Y.C. and Y.H.; software Y.C., Y.H. and C.T.; validation, R.X., Y.C., Y.G. and W.Y.; formal analysis, R.X., C.T., X.J., Y.G. and W.Y.; investigation, R.X., Y.C., Y.H., X.J. and Z.W.; resources, Y.C., H.L. and H.J.; data curation, Y.H., C.T., X.J. and Z.W.; writing—original draft preparation, R.X. and Y.C.; writing—review and editing, R.X., Y.C., Y.H., C.T., X.J., Y.G., W.Y., Z.W., W.S., H.L. and H.J.; visualization, Y.C., Y.H., C.T., W.S., Z.W. and H.L.; supervision, X.J., Y.G., W.Y. and H.J.; project administration, R.X., C.T. and H.J.; funding acquisition, Y.C. and H.J. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the National Key Research and Development Program of China (2025YFD1800402).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: All data in this study are included in the article and Supplementary Materials.

Conflicts of Interest: The authors declare no conflicts of interest.

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